

Editorial: Problem F - Optimal Mirroring

Setter: Riverxia

Problem Overview

Touko wants to maintain the same relative order as a fixed permutation a for q different intervals. Yui needs to find the minimum number of edges in a DAG G_k such that its satisfying permutations are exactly those that match Touko's sister's relative order on the first k intervals. This is equivalent to finding the size of the **transitive reduction** of the partial order defined by the union of constraints from the k intervals.

Solution Sketch

Key Observations

1. An edge (u, v) is required in the transitive reduction if and only if $u < v$ in the partial order and there is no path of length ≥ 2 from u to v .
2. For a single interval $[l, r]$, the minimal set of edges is simply $x \rightarrow y$ for all pairs (x, y) such that a_x and a_y are **consecutive values** in the sorted sub-permutation $a_l, \dots, a_r$. There are exactly $(r - l)$ such edges.
3. When multiple intervals overlap, an edge $u \rightarrow v$ from one interval might be rendered redundant by a path in another.

The Algorithm

A crucial insight from the original problem (CF 1508F) is that we only need to consider “candidate edges.” These are edges (u, v) such that u and v are adjacent in value in *some* range.

Using Mo's algorithm, we can maintain the set of edges where the values are adjacent. As we move the pointers l and r , we add and remove edges. Each time an edge (u, v) is formed, we track the range of k for which it exists.

However, many of these edges are redundant. Lemma 2 from the tutorial states that for any vertex u , there is at most one “right edge” $u \rightarrow u_r$ and at most one “left edge” $u \rightarrow u_l$ in the transitive reduction at any time.

We can identify these candidate edges by running Mo's algorithm. There are $O(N\sqrt{Q})$ such candidate edges. For each candidate edge, we determine the time interval $[t_{start}, t_{end}]$ during which it is the “best” candidate and not “hidden” by a path of length 2.

Complexity Analysis

- **Time Complexity:** $O(N\sqrt{Q} \log N)$ or $O(N\sqrt{Q})$ depending on the implementation of the candidate edge tracking. Given the 7s time limit, a Mo's algorithm approach is feasible.

- **Space Complexity:** $O(N + Q)$ to store the permutation, queries, and candidate edge info.

Bloom into You Connection

Touko's struggle is about the pressure of maintaining a specific relative order (her sister's legacy). Yuu's task of finding the *minimum* number of edges represents her finding the most fundamental truths about Touko—stripping away the redundant layers of the performance until only the essential constraints remain.